THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 3 & 4: **Metals and Non-metals, Carbon and its Compounds** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Magnesium reacts with hot water to produce: 1

(a) magnesium chloride **(b)** only hydrogen
(c) magnesium oxide and oxygen **(d)** magnesium hydroxide and hydrogen

2. Student A says zinc oxide is amphoteric; Student B says it is acidic only. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

3. Which gas is evolved when a metal reacts with dilute HCl? 1

(a) chlorine **(b)** carbon dioxide **(c)** oxygen **(d)** hydrogen

4. The impurities present in an ore are called: 1

(a) slag **(b)** anode mud **(c)** flux **(d)** gangue

5. The metal at the top of the reactivity series (among these) is: 1

(a) gold **(b)** zinc **(c)** potassium **(d)** copper

6. The general formula of alkynes is: 1

(a) C_nH_{2n-2} **(b)** C_nH_n **(c)** C_nH_{2n+2} **(d)** C_nH_{2n}

7. For a mass difference between successive homologues, Student A says 14 u; Student B says 12 u. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

8. Vinegar is a dilute solution of: 1

(a) hydrochloric acid **(b)** methanoic acid
(c) ethanol **(d)** ethanoic acid

9. Substitution of hydrogen in methane by chlorine requires: 1

(a) conc. sulphuric acid **(b)** sunlight
(c) nickel catalyst **(d)** darkness

10. The IUPAC name of $CH_3CH_2CH_2OH$ is: 1

(a) propanoic acid **(b)** propanol **(c)** propanone **(d)** propanal

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is meant by 'anode mud'? 2

12. A student says that aluminium is not used for cooking utensils because it reacts easily. Justify or refute. 2

13. What happens when ethanol is heated with excess conc. H_2SO_4 at 443 K? 2

14. A student says that ethene and ethane both belong to the same homologous series. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Write two differences between metals and non-metals on the basis of chemical properties. 3

16. An organic compound A ($C_2H_4O_2$) turns blue litmus red and reacts with ethanol to give a fruity smell. Identify A and write the reaction. Name the product. 3

17. What are the steps in the extraction of a metal from its ore? Name them in order. 3

OR

Why is iron galvanised? What happens if the zinc coating is scratched? 3

18. Draw the structures of 3

(i) ethanoic acid

(ii) propanone

(iii) chloropropane.

OR

Why does carbon not form C^{4+} or C^{4-} ions? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Write the electronic configuration of Na (11) and Cl (17). 5

(ii) Show the formation of NaCl by electron transfer.

(iii) Give two properties of NaCl due to its ionic nature.

OR

(i) Explain the reaction of metals with 5

(a) oxygen

(b) water

(c) dilute acids, with one equation each.

(ii) Which metal does not react with oxygen even at high temperature?

20. (i) What is catenation? 5

(ii) Why does carbon show it more than silicon?

(iii) Draw the structures of a straight chain, branched chain and ring compound of six carbon atoms.

OR

(i) Describe the chemical properties of ethanol (reaction with Na, dehydration). 5

(ii) Harmful effects of alcohol.

(iii) What is the use of ethanol as a fuel?

END OF QUESTION PAPER — **Total: 40 Marks**